Secure Cloud Network Architecture: Guided Projects

Five projects that build on each other, each ending with deliverables you can put in a portfolio. AWS terms are used, with Azure/GCP equivalents noted where they differ.

Project 1: Secure Landing Zone and Network Foundation

Goal: Build a multi-tier, segmented network from scratch.

Steps

- Design the IP plan (CIDR allocation for prod, dev, shared services, with room to grow)
- Build a VPC/VNet with public, private-app, and private-data subnets across 2-3 availability zones
- Configure route tables, NAT gateways, security groups, and NACLs (Azure: NSGs)
- Add VPC endpoints/Private Link so private workloads reach cloud services without touching the internet

Deliverables

- Network diagram (draw.io or Lucidchart)
- Terraform code for the full network
- IP addressing and segmentation document
- Test evidence showing blocked and allowed traffic paths

Project 2: Hub-and-Spoke with Centralized Inspection

Goal: Connect multiple environments through a controlled hub.

Steps

- Deploy Transit Gateway (Azure: Virtual WAN/hub VNet) with spoke VPCs
- Add a central inspection VPC with a firewall (AWS Network Firewall, Azure Firewall, or a Palo Alto/Fortinet appliance)
- Force east-west and egress traffic through the hub
- Write firewall rules: domain allowlists for egress, IPS signatures, deny-by-default between spokes

Deliverables

- Hub-and-spoke architecture diagram with traffic flow numbering

- Firewall rule matrix (source, destination, port, justification)
- Terraform modules for the hub and spokes
- Validation report using test instances

Project 3: Hybrid Connectivity and Zero Trust Access

Goal: Securely connect on-premises users and sites, and replace flat VPN access.

Steps

- Set up Site-to-Site VPN (IPsec/IKEv2) using a simulated on-prem network (pfSense or a second VPC)
- Add Client VPN or a ZTNA approach (Cloudflare Access, Tailscale, AWS Verified Access)
- Integrate SSO/MFA through an identity provider (Entra ID, Okta, or Keycloak)
- Replace SSH/RDP exposure with Session Manager, Bastion, or Identity-Aware Proxy

Deliverables

- Connectivity design document with encryption parameters
- Access policy matrix (who can reach what, under what conditions)
- Demo showing no inbound management ports are open
- Failover test results (tunnel redundancy)

Project 4: Edge Security for a Public Application

Goal: Protect an internet-facing web app end to end.

Steps

- Front the app with CDN, WAF (OWASP managed rules plus custom rate limits), and DDoS protection
- Configure TLS 1.2+/1.3, HSTS, and certificate automation
- Place the app behind an ALB/App Gateway with private backend instances
- Add DNS security (DNSSEC where supported, Route 53 Resolver DNS Firewall)
- Run an attack simulation (OWASP ZAP, SQLi/XSS attempts) and tune rules

Deliverables

- Threat model (STRIDE) for the application
- WAF rule set with before/after attack test results
- Architecture diagram with trust boundaries marked

- TLS configuration audit (SSL Labs grade A or better)

Project 5: Monitoring, Detection, and Compliance Automation

Goal: Make the network observable and continuously audited.

Steps

- Enable VPC Flow Logs, DNS logs, firewall logs, and CloudTrail/Activity Logs, and centralize them
- Build detections in a SIEM (Security Hub, Sentinel, Splunk, or the ELK stack): port scans, unusual egress, and policy changes
- Add policy-as-code guardrails (AWS Config, Azure Policy, OPA/Checkov in CI) to block insecure changes such as open 0.0.0.0/0 rules
- Run an incident drill: simulate a compromised instance and contain it with automated isolation

Deliverables

- Dashboards and alert rules
- CI pipeline that fails on insecure Terraform
- Mapping of controls to CIS Benchmarks or NIST 800-53
- Incident response runbook plus drill write-up

Capstone: Secure Multi-Tier Architecture

Combine all five projects into one reference architecture with:

- A full design document covering requirements, decisions, and trade-offs
- Complete IaC repository with CI/CD
- Security test report and residual risk register
- A 10-minute architecture walkthrough presentation

Suggested Timeline and Tooling

- **Timeline:** about 2 weeks per project, 10-12 weeks including the capstone
- **Tools:** Terraform, draw.io, Wireshark, nmap, OWASP ZAP, Checkov, pfSense
- **Cost tip:** use free tiers and tear down with `terraform destroy` after each session. NAT gateways and firewalls are the main cost drivers.